CoconutRisk AI — C3 Dataset Quality Report

DATA ORIGIN: 100% SYNTHETIC. Not observed Sri Lankan meteorological data, not mapped coconut plantations, not measured crop stress.

Automated structural validation: PASS

This report certifies consistency of the generated simulation files only. No real-world model skill or climatological accuracy can be inferred.

Check	Verified result
Daily observations	525,952 synthetic records
Historical monthly records	17,280 (1981–2025)
ML forecasting rows	9,600 (2001–2025)
Example Sri Lankan locations	32 illustrative site points
Primary target	SPI-3 at 1/3/6 months ahead
Reference period	Fixed synthetic 1981–2010 climatology
Provenance	Every exported value has is_synthetic=1
Temporal leakage	Labels and splits checked per target horizon
Consistency	Daily rainfall totals match monthly aggregates
Spatial caveat	Demo GeoJSON cells are not verified land-use polygons

Risk categories (simulation only)

Risk band	ML historical rows
Critical	1107
High	1228
Low	5584
Medium	1681

Baselines evaluated on simulated test periods

Horizon	Model	MAE	RMSE
1	Persistence_SPI3	0.50757	0.65179
1	Seasonal_Climatology_SPI3	0.84102	1.05985

Horizon	Model	MAE	RMSE
3	Persistence_SPI3	0.95851	1.19679
3	Seasonal_Climatology_SPI3	0.84102	1.05985
6	Persistence_SPI3	1.13805	1.40197
6	Seasonal_Climatology_SPI3	0.84102	1.05985

Limitations and next steps

Use this package to implement and exercise model-training scripts, APIs and UI. Deep-learning models, live weather feeds and a running full-stack application are NOT included as completed products. Validate and retrain on authentic CHIRPS/ERA5/CRI records before any real-world assessment. A fixed 1981–2010 synthetic gamma reference is used only to test reproducible index calculations.